ATENOLOL

Renal dose adjustment

Do not stop suddenly

CHEMICAL NAME 2-{4-[(2RS)-2-hydroxy-3-(isopropylamino)propoxy]phenyl}acetamide

CLASS Cardioselective (β1) adrenoceptor blocker (hydrophilic)

TRADE NAMES Tenormin · Blokium · Tenoretic (with chlortalidone)

THERAPEUTIC USES / INDICATIONS

- Hypertension (no longer first-line alone).
- Angina pectoris.
- Arrhythmias (SVT, AF rate control).
- Early management of acute MI.

PHARMACODYNAMICS

- Selective **β1 blockade** → ↓ HR, contractility, cardiac output, renin release.
- Less bronchospasm than non-selective agents at low doses.

ADMINISTRATION

- **PO** once daily, same time each day.
- **IV** slowly (1 mg/min) for arrhythmias/MI (where available).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Fatigue, cold extremities.
- Bradycardia.
- GI upset.

ADVERSE EFFECTS (SERIOUS)

- Heart block, severe bradycardia.
- Bronchospasm (high doses, asthma).
- Worsening heart failure.
- Masked hypoglycaemia; withdrawal angina.

DOSAGES

- **Hypertension:** 25–50 mg once daily (max 100 mg).
- **Angina:** 100 mg once daily or 50 mg twice daily.
- **Arrhythmias:** 50–100 mg daily.
- **CrCl 15–35:** max 50 mg/day; **<15:** max 25 mg/day.

CONTRAINDICATIONS & CAUTIONS

- 2nd/3rd-degree heart block, sick sinus syndrome, cardiogenic shock.
- Uncontrolled heart failure.
- Severe bradycardia; severe peripheral arterial disease.
- **Caution:** asthma, diabetes, renal impairment.

KEY INTERACTIONS

- **Verapamil, diltiazem:** bradycardia, heart block.
- **Clonidine:** rebound hypertension.
- **Insulin, sulfonylureas:** masked hypoglycaemia.

PHARMACOKINETICS

- **Onset:** 1 h. **Peak:** 2–4 h. **Duration:** 24 h.
- Hydrophilic — little CNS penetration; not metabolised.
- Excreted unchanged in urine. **t½:** 6–7 h (prolonged in renal failure).

NURSING MANAGEMENT

- Check apical pulse and BP before giving.
- Adjust dose for renal function.
- Monitor for wheeze, oedema, weight gain.
- Monitor glucose in diabetics.
- Taper on discontinuation.

PATIENT EDUCATION

- Take at the same time each day.
- Do not stop suddenly.
- Report slow pulse, dizziness, breathlessness.
- Rise slowly.

KEY POINTS

- β1-selective, hydrophilic, renally excreted.
- Reduce dose in renal impairment.
- Fewer CNS effects.
- Taper to stop.

SOURCES CHECKED **[296]** Atenolol 100 mg film-coated tablets (UK SmPC) **[297]** Atenolol, oral (US label) **[298]** Atenolol, oral (US label, Proficient Rx LP)

Full references p. 432–457